



A survey of MRI-based brain tissue segmentation using deep learning

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Received: 18 January 2024 / Accepted: 23 August 2024 / Published online: 5 December 2024
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Abstract

Segmentation of brain tissue from MR images provides detailed quantitative brain analysis for accurate diagnosis, detection, and classification of brain diseases, and plays an important role in neuroimaging research and clinical environments. Recently, a plethora of deep learning-based approaches have been employed to achieve brain tissue segmentation in fetuses, infants, and adults with impressive outcomes. However, owing to the existence of noise, motion artifacts, and edge blurriness in MR images, automatically segmenting brain tissue accurately from MR images is still a very challenging task. This survey examines both deep learning and MRI, providing an overview of the latest advances in fetal, infant, and adult brain tissue segmentation techniques based on deep learning. It includes the performance and quantitative analysis of the state-of-the-art methods. Over 100 scientific papers covering various technical aspects, including network architecture, prior knowledge, and attention mechanisms, were reviewed and analyzed. This article also comprehensively discusses these technologies and their potential applications in the future. Brain tissue segmentation provides detailed quantitative brain analysis for accurate diagnosis, detection, and classification of brain diseases, and plays an important role in neuroimaging research and clinical environments.

Keywords Brain tissue segmentation · Survey · Fetal · Infant · Adult · Deep learning · Attention mechanism · Network structure · Prior knowledge

Keywords المسبقة الكلمات الرئيسية : تجزئة الدماغ · والتحقيق · الجنين · الطفل · الكبار · والتعلم العميق · الانتباه إلى آلية · هيكل الشبكة · المعرفة

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Introduction

Magnetic resonance imaging (MRI) is a safe and noninvasive imaging modality that provides high-resolution brain images with various contrasts. Owing to its capabilities, it has been widely used in the examination of neurological disorders. MRI offers a powerful non-invasive method for studying brain structures and function within the body, which has sparked interest in evaluating the development of infant and adult brains. This interest stems from the fact that MRI allows noninvasive and safe examination of the brain, providing multiple cross-sectional views with different contrasts [1]. Compared with traditional methods, deep learning-based automatic segmentation methods have significant advantages in terms of accuracy and processing efficiency. The deep learning methods can accurately extract regions of interest in the brain and differentiate them from normal brain tissue, thus providing a more accurate quantitative analysis. The segmentation of gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF) within the brain is crucial in various clinical and neuroscience studies. This deep learning-based segmentation enables visualization and analysis of anatomical structures, facilitates surgical planning, and supports image-guided interventions [2, 3].

Recently, deep learning has gained widespread popularity for medical image segmentation. Compared with the traditional medical image segmentation methods, the deep learning-based methods exhibit greater generalization performances, automatic feature extractions, and parallel working abilities [4]. The deep learning-based methods have also achieved great strides in brain tissue segmentation, including for fetuses, infants and adults. Despite the complexity and intensity variations of brain MR images, brain tissue segmentation continues to face many challenges. Accurate segmentation via MR images has become a challenging task for diagnosing and treating patients more effectively. However, achieving tissue segmentation is challenging due to the complex structures and intensity variations in the brain MR images. To diagnose and treat patients more effectively, accurate segmentation via MRI remains a challenging task [5, 6]. Currently, there is a lack of sufficient training data needed for accurate segmentation, which may lead to the inadequate generalizability of the segmentation model. Additionally, the complexity and diverse structures of brain tissues pose many difficulties for automatic segmentation. In particular, various factors, such as different magnetic resonance imaging parameters, noise and artifacts, can influence the accuracy of the segmentation results. To address these challenges, further algorithmic improvements are necessary for researchers.

Although many studies have been devoted to developing new ideas for deep learning-based brain tissue segmentation tasks, these ideas have also introduced new problems that remain unsolved. An extensive literature search over

research from the past 10 years was conducted on the basis of keywords, via PubMed and Google Scholar. Unlike other literature reviews in the field of brain segmentation, we focused on deep learning-based MR image segmentation algorithms for normal fetal, infant and adult brain tissue segmentation [6–9]. Hence, the keywords used for selecting studies were “brain tissue segmentation,” “deep learning,” “adult brain tissue segmentation,” “infant brain tissue segmentation,” and “fetal brain tissue segmentation.” First, this paper briefly introduces commonly used databases for brain tissue segmentation. To explore and promote the development of deep learning-based methods in brain tissue segmentation, we review more than 100 existing segmentation methods, providing solutions for clinical diagnosis and medical applications. Finally, we discuss new strategies to improve the performance and interpretability of deep learning-based segmentation networks. The contributions of this review are as follows:

- (1) This survey delves into the latest technology of using deep learning for fetal, infant, and adult brain tissue segmentation in MR images, providing important technical support for medical image analysis and neuroscience research.
- (2) This survey can help clinicians more accurately diagnose and evaluate brain diseases, such as tumors and cerebral infarction, and determine the appropriate treatment plans.
- (3) This survey presents novel research avenues for deep learning-based brain tissue segmentation for researchers. It also plays an important role in neuroscience research, helping scientists better understand the structure and function of the brain.

The rest of the paper is structured as follows: Sect. “[Datasets for brain tissue segmentation](#)” provides an overview of commonly used publicly available brain segmentation datasets and challenge provided datasets. Sect. “[Methods of segmentation](#)” presents various deep learning-based segmentation methods for fetal, infant and adult brains used in the literature, and Sect. “[Discussion](#)” identifies potential future research directions for brain tissue segmentation. Finally, Sect. “[Conclusion](#)” offers concluding remarks on the review.

Datasets for brain tissue segmentation

Numerous institutions and researchers have released datasets to advance the development and promotion of automatic brain tissue segmentation methods. Table 1 provides an overview of public and challenge datasets on deep learning-based brain tissue segmentation.

Table 1 overview of public and challenge datasets on deep learning-based brain segmentation

Dataset	Modality	Target	Total number of images
BrainWeb	T1w, T2w, PDw	Adult	20
IBSR18	T1w	Adult	20
IBSR20	T1w	Adult	20
CANDI	T1w	Adult	103
OASIS-1	T1w	Adult	416
dHCP	T1w, T2w, rs-fMRI, dMRI	Infant and adult	2469 infants and 250 adults
MRBrainS13	T1w, T1w inversion recovery, and FLAIR	Adult	20
MRBrainS18	T1w, T1w inversion recovery, and FLAIR	Adult	30
iSeg-2017	T1w, T2w	Infant	23
iSeg-2019	T1w, T2w	Infant	23
MICCAI 2012 challenge	T1w	Adult	35
DLBS	T1w	Adult	315
SALD	T1w	Adult	494
IXI	T1w, T2w, PD, MRA, DTI	Adult	600
LPBA40	T1w	Adult	40
ABIDE	T1w	Adult	2264
FeTA 2021	T2w	Fetal	80

BrainWeb [10, 11] is a simulated brain dataset. It contains a set of real MR images generated by an MRI simulator. We can obtain different types of MR images by setting the modality, slice thickness, noise level and intensity nonuniformity (INU) field. Twenty normal anatomical models are now available. Figure 1 shows examples of T1-weighted

(T1w), T2-weighted (T2w), PD-weighted (PDw) and the corresponding ground truth images from the BrainWeb dataset.

The Internet Brain Segmentation Repository (IBSR) [12] provides T1w brain images and the corresponding manually guided expert segmentation results, including GM, WM, and CSF. It consists of the IBSR18 and IBSR20 datasets.

The Child and Adolescent NeuroDevelopment Initiative (CANDI) [13, 14] contains 103 T1w brain images and the corresponding labels of 39 structures with manually guided expert segmentation results. Wang et al. assigned each of the 39 structures to the background, CSF, GM, and WM to obtain the segmentation results of four brain tissues [15].

The Open Access Series of Imaging Studies (OASIS)-1 [16] consists of a cross-sectional collection of 416 subjects aged 18 to 96. For each subject, 3 or 4 individual T1w brain images obtained in single scan sessions are included. It also includes manually-guided expert segmentation results for GM, WM, and CSF.

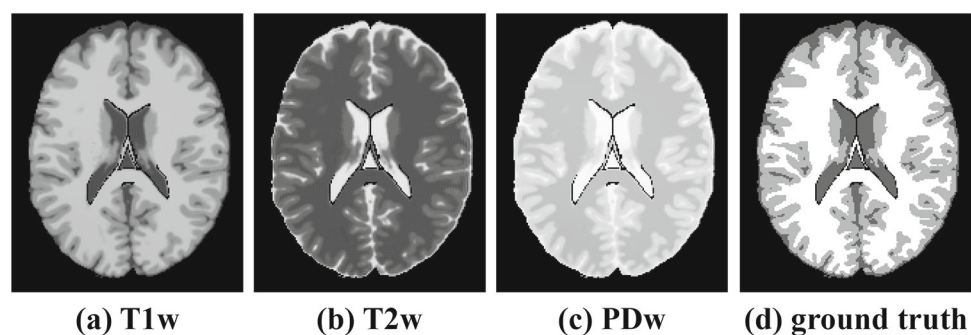
The Developing Human Connectome Project (dHCP) [17] has released data three times, including fetal and infant data. It contains anatomical images (T1w and T2w), resting state functional MRIs (rs-fMRI), diffusion MRIs (dMRI), and so on [18].

The MRBrainS13 [19] dataset was provided by an international conference on medical image computing and computer-assisted intervention (MICCAI) brain tissue segmentation challenge in 2013. Its purpose is to compare (semi-) automatic algorithms for segmentation of GM, WM, and CSF on multiple sequence (T1w, T1w inversion recovery, and FLAIR) brain MR images.

The MRBrainS18 [19] dataset was provided by an international conference on the MICCAI brain tissue segmentation challenge in 2018. Unlike the MRBrainS13 dataset, this dataset not only compares the performance of different segmentation methods, but also assesses the effect of (large) pathology on segmentation and volumetry.

The MICCAI 2017 Grand Challenge on 6-month infant brain MRI segmentation (iSeg-2017) [20] dataset provides 10 subjects with different image modalities (T1w and T2w) for training and 13 subjects for testing. The purpose of iSeg-2017

Fig. 1 The different MRI weighted images and ground truth are from the BrainWeb dataset. From left to right: T1w, T2w, PDw, and ground truth image. In (d), white is WM, light gray is GM, and dark gray is CSF



is to encourage the development of automatic segmentation algorithms for 6-month-old infant brain images.

The iSeg-2019 [21] dataset provides T1w and T2w images of 10 infant subjects for training, 13 subjects for validation, and 16 subjects from 3 sites for testing. This challenge aims to promote the use of an automatic segmentation algorithm for 6-month infant brain MR images at multiple sites. Both iSeg-2017 and iSeg-2019 provide manual segmentation results for GM, WM, and CSF.

The MICCAI 2012 challenge [22] dataset includes 35 brain MR images obtained from the OASIS dataset.

The Dallas Lifespan Brain Study (DLBS) [23] contains 315 anatomical MR images. This dataset focuses particularly on the early stages of healthy brain progression toward Alzheimer's disease (AD).

The Southwest University Adult Lifespan Dataset (SALD) [24] is a multi-model dataset containing sMRI, rs-fMRI, and behavioral data. The aim is to reveal the developmental trajectory of the human brain and understand the changes that occur as we age.

The IXI [25] dataset includes nearly 600 MR images collected from 3 different hospitals. Each subject had T1w, T2w, PDw images, magnetic resonance angiography (MRA) image, and Diffusion-weighted images.

The Autism Brain Imaging Data Exchange (ABIDE) [26] includes two large collections: the ABIDE I and ABIDE II datasets. The data for each dataset came from more than 24 international brain imaging laboratories.

The LONI Probabilistic Brain Atlas (LPBA40) [27] is a collection of brain anatomic region maps. These maps were created via whole-head MR images of 40 human subjects, and each MRI image was meticulously outlined to identify 56 structures in the brain. These structures are located primarily within the cortex.

Methods of segmentation

In recent years, fetal, infant and adult brains have attracted extensive research interest in various countries around the world [28–31]. The utilization of MR images in fetuses aids clinicians in identifying brain abnormalities during the early stages of development. Brain tissue segmentation of fetal images is the foundation of volumetric and morphologic analysis [32]. The segmentation of infant brain tissue from MR images is a crucial step in studying early brain development in infant health and disease. This process holds great significance as it allows for a better understanding of the brain's growth and can aid in identifying potential health issues at an early stage. Furthermore, segmentation of adult brain tissue is an important tool for studying neurodegenerative diseases [33]. However, owing to significant differences in brain size, appearance, and shape among fetuses, infants,

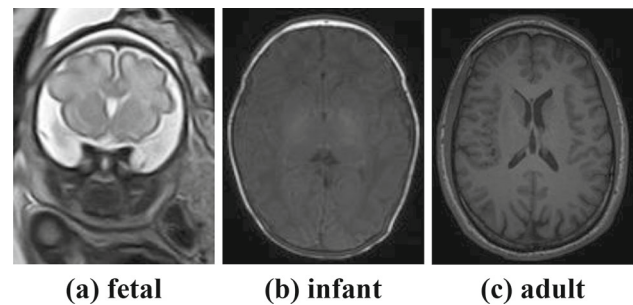


Fig. 2 The brain MRI images shows different stages. **a** fetal; **b** infant; **c** adult

and adults, it is necessary to design different segmentation algorithms for different age groups. In the case of fetuses, image distortions can occur due to artifacts caused by factors such as the autonomous breathing of the mother, amniotic fluid, and placenta. Additionally, MR images of newborns are usually acquired during deep sleep, leading to image blurring and decreased contrast, which poses challenges for segmentation algorithms. Moreover, there is often a scarcity of available fetal and newborn MRI datasets for training and validation, compared with the amount of MRI data available for adults. This results in smaller dataset sizes for brain segmentation tasks, making the training and evaluation of segmentation algorithms more challenging. Owing to the lack of large-scale annotated data, establishing accurate and reliable segmentation models is difficult. Hence, most existing deep learning-based segmentation methods for adult brains are not applicable to fetal and infant brains [29]. Figure 2 presents fetal and infant brain MR images from Taian Maternal and Child Health Hospital and adult brain MR images from public datasets. This survey summarizes the deep learning-based brain tissue segmentation methods for fetuses, infants and adults, to gain some insight into this field.

Segmentation of fetal brain tissue

By using MR images of fetuses, doctors can detect abnormalities in fetal brains during their early stages of development. For example, premature babies are at high risk for cerebral palsy. In fact, problematic cases can be traced back to the fetal stage [34]. Before performing brain morphology and volume analysis, it is necessary to segment the brain tissue [35]. Table 2 provides an overview of the datasets, deep learning-based models, and mean Dice similarity coefficient (DSC) used for fetal brain tissue segmentation.

Convolutional neural networks (CNNs) have been widely used in fields such as denoising, segmentation, registration, and classification. It achieves better performance than traditional algorithms. CNN have achieved great success in fetal brain tissue segmentation. Some researchers have applied or

Table 2 A brief summary of the deep learn-based methods for fetal brain tissue segmentation

Paper	Architecture	Dataset	Mean DSC
Fetit et al. [36]	DeepMedic 0.7.0	dHCP	0.8186
Li et al. [37]	CAS-Net	dHCP	0.8520
Payette et al. [38]	U-net	University Childrens Hospital Zurich	0.7900
Khalili et al. [32]	Two U-net	University Medical Center (UMC) Utrecht	0.8840
Peng et al. [35]	nnU-Net	FeTA 2021	0.8420
Zhao et al. [39]	3D U-Net	Discovery MR450 GE Healthcare	0.8819
Pei et al. [40]	PETS-Nets	Hospital	0.9153
Karimi et al. [41]	nnU-Net	Hospital	0.8720
Huang et al. [42]	CNN	University Childrens Hospital Zurich	0.8379
Wen et al. [43]	ExSwin-Unet	FeTA 2021	0.9010
Lee et al. [44]	3D UX-Net	FeTA 2021	0.8740
Zhang et al. [45]	U-Net	Siemens	0.9239
Sun et al. [46]	MSMHA-CNN	Hospital	0.9478

proposed many CNN-based segmentation models for fetal brain images, such as Fetit et al. [36] and Li et al. [37]. Specifically, Fetit et al. focused on addressing the lack of extensive/publicly available annotations in fetal brain segmentation via a two-stage CNN model. Both CNN models utilized DeepMedic v0.7.0 [47]. Li et al. proposed a multi-task learning model for fetal brain tissue segmentation, called CAS-Net. It can simultaneously generate conditional atlases and predict brain tissue segmentation maps. The anatomical priors provided by conditional atlases can constrain segmentation connectivity [37].

As a convolutional network structure for image segmentation, the U-Net model outperforms the current best methods in medical image segmentation [48]. Figure 3 shows the structure of a U-Net network. Scholars have proposed many improved and derived segmentation models based on the U-Net model, such as those of Payette et al. [38], Khalili et al. [32], Zhao et al. [39], Peng et al. [35] and Pei et al. [40]. Payette et al. focused on the problem of noise labels, used noise labels to train the U-Net network, and introduced transfer learning to further improve the segmentation

results. Khalili et al. designed a two-stage segmentation task using two identical U-Net networks. The first stage obtains the intracranial volume, and the second stage obtains the segmented different brain tissues. Isensee et al. proposed a self-configuring segmentation method, called nnU-Net [49]. It can improve the performance and generality of the model by automatically configuring itself. Peng et al. applied nnU-Net to fetal brain tissue segmentation and ranked third in the FETA 2021 challenge [35]. Pei et al. proposed a novel multi-task learning framework that adopts a coarse-to-fine strategy to jointly learn the pose estimation parameters for motion correction and a tissue segmentation map of each slice in fetal MR images. Experimental results on different datasets show that the proposed model can obtain superior tissue segmentation results [40]. Karimi et al. used the nnU-Net model to segment fetal brain tissue from noisy annotations. On manually labeled images, the proposed model achieves superior segmentation accuracy [41].

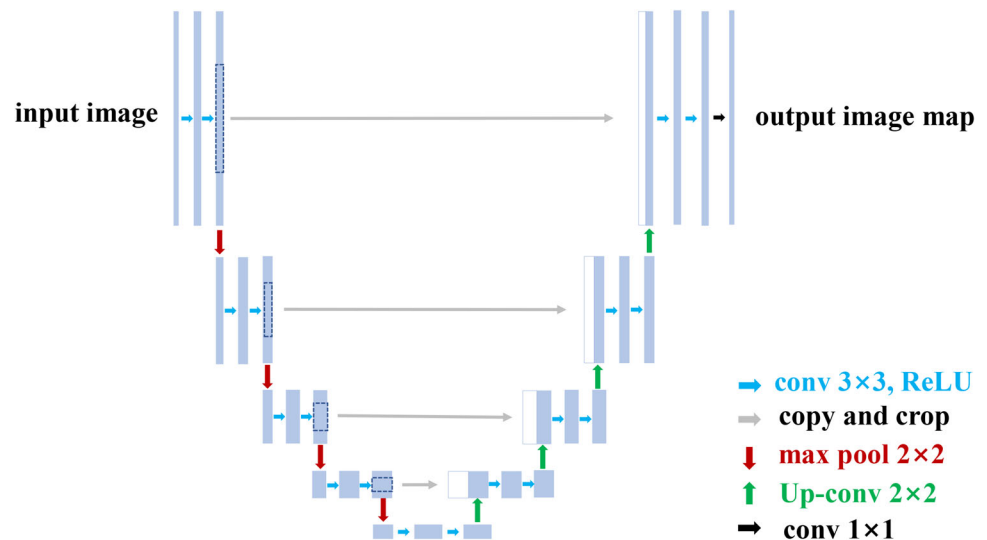
Currently, attention mechanisms have been used for effective segmentation [42–46, 50–53]. Transformation-based models combined with attention mechanisms have achieved great progress in the image segmentation field [50, 51]. Figure 4 shows the Transformer module architecture. Huang et al. designed a new deep learning-based fetal brain tissue segmentation model by introducing a contextual Transformer block (CoT-Block) and a hybrid dilated convolution module [42]. Segmentation experiments on 80 fetal brain MR images show that the proposed model has a good segmentation performance. Wen et al. [43], Lee et al. [44], and Zhang et al. [45] improved the Transformer module and proposed their respective fetal brain tissue segmentation models, both of which enhanced the segmentation performance of the model. Payette et al. organized the 2021 Fetal Tissue Annotation (FeTA) challenge and performed a review analysis of the 21 results that participated in the competition [52]. On the basis of the success of FeTA 2021, Payette et al. organized FeTA 2022 to develop segmentation methods across different sites [53].

Segmentation of infant brain tissue

According to Li et al. [29], the initial years after birth are the most vital and dynamic phase of postnatal brain development. Precise segmentation of brain tissue in MR images of infants is a crucial step in examining early brain development in both healthy and diseased states [54]. However, this is very challenging because of the lower tissue contrast in early-stage brains [55]. Table 3 summarizes the models, datasets, and mean DSCs of the deep learning-based methods for infant brain tissue segmentation.

The rapid development of deep learning has brought great opportunities and potential for the development of infant brain tissue segmentation [78]. The field of image processing

Fig. 3 The structure of U-net (Ronneberger et al., 2015). It contains the encoder and decoder paths, with skip connections between corresponding layers in the encoder and decoder parts of the network. This enables the network to retain both local and global information during the segmentation process. The encoder and decoder paths were also composed of four level



widely utilizes CNNs because of their due to powerful feature extraction ability [79]. Owing to the high spatial resolution of T1w images, several studies have focused on the use of T1 images for infant brain tissue segmentation [56], such as Bui et al. [56], Fan et al. [57], Ding et al. [58], etc. Specifically, Bui et al. introduced a multi-task learning strategy to simultaneously learn geodesic distance regression and brain tissue segmentation from T1w image. Fan et al. improved the UNet model for infant brain tissue segmentation [57]. A multi-scale dilated convolution module is introduced in the encoder path for complete feature extraction, and a multi-branch attention module is added to the decoder for feature reconstruction. Ding et al. evaluated modified LiviaNet [80] and HyperDense-Net [81] on the dHCP neonatal dataset. Experiments show that both networks are able to segment neonatal brains according to previously reported performance.

In the first year of life, brain MRI consists of three phases: (1) the infantile phase (≤ 5 months), (2) the isointense phase (6–8 months), and (3) the early adult-like phase (≥ 9 months) [20]. The voxel-intensity iso-intensity phases of GM and WM overlap to a large extent, which makes brain tissue segmentation highly challenging. Many studies have focused on segmenting MR images of neonatal and early adult-like brains. Wang et al. organized the iSeg-2017 challenge to improve automatic segmentation algorithms for 6-month infant brain MR images [20]. The dataset contains two types of brain MR images: T1w and T2w. Scholars have proposed many automatic algorithms for the iSeg-2017 dataset. Zeng et al. designed context-guided, multi-stream fully convolutional networks (FCN) for multi-modal infant brain tissue segmentation [59]. Dolz et al. [60], Dolz et al. [61], Bui et al. [63], Hashemi et al. [64] and Qamar et al. [82] all proposed infant brain tissue segmentation methods by introducing dense connections. Dense connections improve the data information flow between layers and are conducive

to the reuse of features, ultimately leading to improved segmentation performance of the model. Li et al. applied deep learning to the sub-regions defined by the registration-based segmentation method, which can capture more features about these sub-regions and obtain finer segmentation results [62]. Ding et al. combined fuzzy adjacency information in a deep learning-based model for 6-month infant brain tissue segmentation [65]. Fuzzy features can represent image uncertainty and improve the generalization ability of the model. Khaled et al. applied Generative Adversarial Network (GAN) to infant brain tissue segmentation [66]. Dénes-Fazakas et al. also proposed a derived U-Net model for brain tissue segmentation [67]. In order to learn the complex non-linear complementary relationships between images of different modalities, Zhuang et al. [68] proposed a series of learning modules and conducted experiments on both adult and infant datasets. The experimental results show that the proposed model achieves state-of-the-art segmentation performances.

The common limitations of the algorithms proposed using the iSeg 2017 dataset may not apply to images acquired from different sites, scanners or imaging protocols. Hence, Wang et al. organized iSeg 2019 to facilitate automatic segmentation algorithms for brain MRIs in 6-month infants from multiple sites [21]. The organizers then assessed the 8 top-ranked submitted algorithms, and noted out the limitations of these algorithms [21]. In addition, some scholars have proposed deep learning-based segmentation algorithms for the iSeg 2017 and iSeg 2019 datasets. Wei et al. proposed a 3D U-Net for infant tissue segmentation, which combines the attention mechanism module and the pyramid structure module [69]. To extract multi-dimensional features between MR images, Luan et al. proposed a convolutional mechanism for feature extraction, called Rubik convolution. Furthermore, the authors proposed an efficient 2D convolutional network, namely, Rubik-Net, for brain tissue segmentation

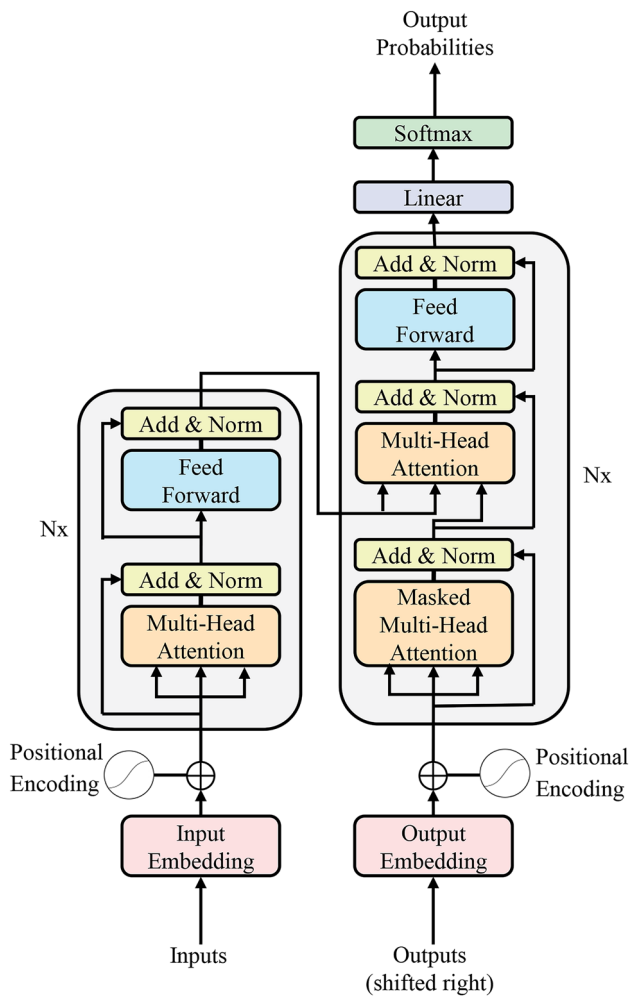


Fig. 4 The structure of Transformer module (Vaswani et al., 2017). It includes encoder (left) and decoder (right). The decoder is an extended encoder. The encoder encrypts the input sequence. It has a multi-head attention layer and a regular feed forward layer. The decoder has two multi-head attention layers and a fully connected feed forward network. Its core innovation lies in the reliance on self-attention mechanisms instead of recurrence or convolution, enabling parallelization and scaling to longer sequences effectively

[70]. Omid et al. presented an unsupervised method to adapt a UNet skull-stripping model trained on adult MR images to work effectively on new-borns. Sun et al. proposed an accurate self-supervised learning framework for multi-site infant cerebellum segmentation [71]. ADU-Net was chosen as the backbone supervised segmentation model. The confidence map generated by the segmentation model is used to train a confidence network that predicts the reliability of the automated segmentation [72]. Chen et al. proposed an ANUBEX (automated neonatal nnU-Net brain extractor) designed with nnU-Net. To improve the generalization of image acquisition parameters, it is trained on large multi-institutional datasets and shows accurate performance in high- and low-resolution MR images [73]. Chen et al. proposed a context-guided,

Table 3 A brief summary of the deep learn-based methods for infant brain tissue segmentation

Paper	Architecture	Dataset	Mean DSC
Bui et al. [56]	3D Dense-Unet	GE, Siemens	0.9790
Fan et al. [57]	AMCNN	the local hospital	0.9277
Ding et al. [58]	LiviaNET, HyperDenseNet	dHCP	0.9367
Zeng et al. [59]	FCN	iSeg 2017	0.9220
Dolz et al. [60]	HyperDenseNet	iSeg 2017	0.9260
Dolz et al. [61]	CNNs	iSeg 2017	0.9267
Li et al. [62]	FCNN	iSeg 2017	0.9260
Bui et al. [63]	3D SkipDenseSeg	iSeg 2017	0.9281
Hashemi et al. [64]	DenseNets	iSeg 2017	0.9310
Qamar et al. [58]	3D-UNet	iSeg 2017	0.9159
Ding et al. [65]	FI-DL-Seg	iSeg 2017	0.9513
Khaled et al. [66]	GAN	iSeg 2017	0.9300
Denes-Fazakas et al. [67]	U-Net	iSeg 2017	0.8947
Zhuang et al. [68]	ACMINet	iSeg 2017	0.9316
Wei et al. [69]	3D U-Net	iSeg 2017, iSeg 2019	0.9310, 0.8487
Luan et al. [70]	Rubk-Net	iSeg 2017, iSeg 2019	0.9253, 0.8347
Omid et al. [71]	3D U-Net	GE	0.9156
Sun et al. [72]	ADU-Net	iSeg 2019	0.9153
Chen et al. [73]	nnU-Net	dHCP	0.9250
Chen et al. [74]	M2A-Unet	BCP	0.9450
Nie et al. [75]	FCN	Siemens	0.8891
Nie et al. [76]	FCNs	Siemens	0.8707
Ayaz et al. [77]	3D Residual U-Net	HCP	0.7902

attention-based, coarse-to-fine deep framework to precisely segment infant subcortical structures. The performance of the proposed model was validated on an infant MR image dataset and an unrelated newborn MR image dataset [74]. Most of the above methods use T1w and T2w images, and some scholars have introduced FA images and proposed different segmentation methods. Nie et al. (2018) proposed a

novel 3D multi-modal FCN for segmenting infant brain MR images [75]. In this model, the innovation is to propose a transformation module and a fusion module. Zhang et al. used a CNN model for infant brain tissue segmentation [45]. They integrate multi-modality from T1w, T2w, and fractional anisotropy (FA) images to significantly improve segmentation performance. However, this method does not sufficiently extract information from multimodal images. To address this issue, Nie et al. designed three identical networks with inputs T1, T2, and FA images [76]. Then, the three different types of features are fused to obtain the final segmentation result. Ayaz et al. proposed a realistic simulation framework by incorporating patient-specific phantoms and Bloch equation-based analytical solutions for fast and accurate MRI simulations [77].

Segmentation of adult brain tissue

Deep learning-based segmentation methods have been extensively studied in adult brain MR images [83]. In the literature on brain tissue segmentation, many successful deep learning-based models have been proposed. Among these methods, the popular segmentation strategies include: (1) changing the structure of the model; (2) adding prior knowledge; and (3) introducing an attention mechanism module. Next, we summarize the segmentation methods for adult brain tissue from different segmentation strategies.

Many models of different structures with superior performance exist in deep learning, such as CNNs, residual networks (ResNets) [84], GAN [85], U-Net [48, 86], V-Net [87], etc. Table 4 briefly summarizes the different network structures proposed in the existing studies. Moeskops et al. verified the capabilities of CNN models in different segmentation tasks [86]. In the same year, the author proposed an automatic segmentation model for MR brain images that is based on a CNN [88]. Kong et al. utilized a CNN to realize automatic brain tissue segmentation [89]. Ito et al. proposed a semi-supervised learning framework to obtain segmented pseudo-labels through registration techniques [90]. Later, dilated convolution, dense connection, and ensemble model strategies were also introduced into brain tissue segmentation algorithms [91–96]. The U-Net structure has been widely used in the field of image segmentation, and many studies have proposed new approaches for brain tissue segmentation based on U-Net [97–109]. Rajchl et al. proposed a brain tissue segmentation tool with a multi-output CNN architecture, which was trained with a large number of data sets and enhanced the robustness of the model [110]. Cullen et al. proposed a fully convolutional brain tissue segmentation model with a convolution-deconvolution structure, which quickly completes the segmentation task compared to the traditional methods [111]. Yan et al. proposed a new

Table 4 A brief summary of adult brain tissue segmentation methods with different network structures

Paper	Architecture	Dataset	Mean DSC
Moeskops et al. [86]	CNN	OASIS	–
Park et al. [87]	V-Net	LPBA40	0.9640
Moeskops et al. [88]	Multi-scale CNN	MRBrainS13	0.8633
Fedorov et al. [91]	MeshNet	MRBrainS13	0.8650
Nguyen et al. [92]	GMMs, CNNs, DNNs	IBSR18	0.8600
Kumar et al. [97]	U-SegNet	IBSR18	0.8974
Rajchl et al. [110]	NeuroNet	UK Biobank	0.9400
Gottapu et al. [93]	DenseNet	IBSR18	0.6730
Cullen et al. [111]	CNN	DLBS	0.9001
Dey et al. [98]	CompNet	OASIS	0.9827
Kolak et al. [94]	3D Dense-U-Net	The University Hospital Brno	0.9885
Ito et al. [90]	DNN	IBSR18	0.9400
Wang et al. [15]	RP-Net	CANDI, IBSR18, IBSR20	0.9070, 0.9049, 0.8496
Mjica-Vargas et al. [99]	CNNs	BrainWeb	0.9310
Yan et al. [112] (2019)	GCNs	BrainWeb, IBSR18	0.9167, 0.8600
Kong et al. [89]	CNNs	private	–
Yogananda et al. [100]	3D Dense-Unets	iTAKL study	0.9017
Ramzan et al. [95]	3D CNN	MRBrainS18	0.9140
Yamanakkanavar et al. [114]	M-net	OASIS, IBSR18	0.9553, 0.8999
Mostapha et al. [101]	3D U-Net	ADNI	0.9037
Wei et al. [115]	CaNes-Net	ADNI	0.8852
Zhang et al. [116]	GSCFN	BrainWeb, IBSR18	0.9200, 0.7933
Woo et al. [102]	U-Net	OASIS	0.9490
Weiss et al. [103]	3D U-Net	MRBrainS18	0.8600
Prajapati et al. [104]	SIP-UNet	OASIS	0.9520

Table 4 (continued)

Paper	Architecture	Dataset	Mean DSC
Niu et al. [105]	P-ResUnet	IBSR18	0.9109
Dayanandaet al. [106]	U-SegNet	OASIS, IBSR18	0.9567, 0.9000
Mecheter et al. [96]	CNNs	Clinical dataset	0.8452
Bernal et al. [117]	FCNN	IBSR18, MIC-CAI2012, iSeg2017	–
Goyal et al. [107]	SegNet	IBSR18	0.8300
Gven et al. [118]	SSimDCL	brain-MRI dataset	0.9940
Basnet et al. [108]	3D DCNN	IBSR18, iSeg2017	0.9315, 0.9315
Mohammadi et al. [109]	CycleFormer	BrainWeb	0.8250

brain tissue segmentation model by combining supervoxels and graph convolutional networks (GCN) [112]. Inspired by M-net [113], Yamanakkanavar et al. proposed a patch-wise M-Net network structure for brain tissue segmentation [114]. The model is able to extract finer features that are beneficial for segmentation. Wei et al. designed a cascaded nested network (CaNes-Net) for brain tissue segmentation [115]. The training data for the model are from 7 T images, whereas the test data are the corresponding 3 T images. Zhang et al. proposed a semi-supervised learning brain tissue segmentation model, named the graph self-construction and fusion network (GSCFN) [116]. Specifically, first, the proposed model uses a graph self-construction (GSC) network to explore the relationships between brain tissues. Second, it uses a multi-graph fusion module to integrate multiple types of features to obtain more accurate segmentation results. Bernal et al. implemented eight deep learning-based models for brain tissue segmentation [117]. The segmentation performance of the proposed model was verified in infant and adult datasets. Güven et al. designed a novel GAN model for brain MR image segmentation, called SSimDCL (Supervised SimDCL) [118]. In four different studies, the proposed model yields satisfactory results.

By considering prior knowledge of human brain structure and function, we can significantly improve the accuracy of brain tissue segmentation [64, 115, 116]. Handcrafted features, morphological information, anatomical knowledge, the volumetric shape, the probabilistic atlas, etc., are commonly used prior knowledge in brain tissue segmentation. Table 5 briefly summarizes the different types of prior knowledge proposed in existing studies. Commonly used prior knowledge in brain tissue segmentation includes handcrafted

features, morphological information, anatomical priors, volumetric shapes, brain atlas, etc. [102–106]. More specifically, we introduced texture features to propose a dual-encoder brain tissue segmentation model, which can provide boundary information to improve segmentation performance in our previous study [102]. Morphological information can be used as prior knowledge to enhance the structural details of the image and improve the segmentation performance of the model [103]. For unpaired training data, Dalca et al. used unpaired segmented images to build an anatomical prior and proposed a generation probability model [104]. Experiments using MR images with over 14,000 scans have demonstrated that anatomical priors enable fast unsupervised segmentation. Mahbod et al. introduced a supervised learning framework with volumetric shape models, which can be used as prior knowledge to help segmentation models learn local adaptive classification rules. The probabilistic atlas is a widely used method for brain tissue segmentation [105]. Dalca et al. integrated the probability atlas as prior knowledge into a deep learning segmentation model, which improved the segmentation ability of the model [106].

With the rapid development of medical imaging equipment, studies have shown that multi-modal images can also be used as a priori knowledge to improve the segmentation ability of the model [96, 102, 117, 134, 135]. In brain tissue segmentation, the commonly used image types are T1w, T1-IR, T2w, T2-FLAIR, PDw, etc. Dolz et al. used T1 and T2 modal images to propose a 3D fully convolutional neural network named hyperDenseNet [78]. This model designs a separate feature extraction path for each modality image, which improves the segmentation performance of the model. Xie et al. used T1w, T2w and PDw images for brain tissue segmentation [120]. This method uses LSTM models to classify feature sequences generated by pixel-wise and superpixel-wise constraints. Several studies have proposed brain tissue segmentation models using three types of T1, T1-IR, and T2-FLAIR images [107, 108, 118–120, 122, 123, 136]. For example, Chen et al. proposed a voxelwise residual network (VoxResNet) that integrates multimodal and multi-level information for brain tissue segmentation [107]. Hou et al. cascaded two 3D CNNs to achieve segmentation from coarse-to-fine brain tissue [118]. Moeskops et al. used multi-scale CNN to achieve can not only brain tissue segmentation, but also white matter hyperintensity (WMH) segmentation [108]. To better exploit and fuse information in multimodal images, Chen et al. proposed a novel deep learning architecture, OctopusNet [129]. Li et al. developed a multimodal aggregation network (MMAN) that can extract complementary information from multimodal MR images for fast and accurate segmentation [120].

The attention mechanism is a widely used tool in deep learning [109, 119, 137, 138]. It can focus on important or interesting information and ignore useless information

Table 5 A brief summary of adult brain tissue segmentation methods with different prior knowledge

Paper	Prior knowledge	Dataset	Mean DSC
Dolz et al. [81]	Multi-modal images	iSeg 2017, MRBrainS 2013	0.9266, 0.9257
Wu et al. [119]	Texture feature	BrainWeb, OASIS, CANDI	0.9556, 0.9704, 0.9638
Ghosal et al. [120]	Morphological Gradient	iSeg 2017, IBSR18	0.9630, 0.9289
Dalca et al. [121]	Anatomical prior	ADNI, OASIS, ABIDE, ADHD200, MCIC, PPMI, HABS, Harvard GSP	–
Mahbod et al. [122]	Shape context	MRBrainS13	0.8534
Dalca et al. [123]	Probabilistic atlas	OASIS, MCIC, PPMI, HABS, GSP, ABIDE, ADHD200	0.9350
Chen et al. [124]	Multi-modal images	MRBrainS 2013	0.8742
Hou et al. [125]	Multi-modal images	MRBrainS 2013	0.8617
Moeskops et al. [126]	Multi-modal images	MRBrainS 2013	0.8686
Li et al. [127]	Multi-modal images	MRBrainS 2018	0.8090
Sun et al. [128]	Multi-modal images	MRBrainS 2018	0.8559
Chen et al. [129]	Multi-modal images	MRBrainS2013, ISLES-2018	0.8292, 0.5790
Li et al. [130]	Multi-modal images	MRBrainS 2013	0.8657
Xie et al. [131]	Multi-modal images	BrainWeb, MRBrainS 2013	0.9866, 0.8702
Luna et al. [132]	Multi-modal images	MRBrainS 2018	0.8550
Sun et al. [133]	Multi-modal images	MRBrainS13, OASIS	0.8709, 0.8482

[139, 140]. Table 5 briefly summarizes the different attention mechanisms proposed in the existing studies. Rao et al. [23], Celik et al. [141], and Lee et al. [142] proposed brain tissue segmentation models by introducing the Transformer module, position attention module (PAM) [143], and convolutional block attention module (CBAM) [144], respectively, and validated the performance of the attention mechanism module [145]. For brain tissue segmentation from multi-modal images, Zhang et al. proposed a multi-modality reconstruction attention (MRA) and Zhuang et al. [146] proposed a multi-modal cross-dimension attention (MCDA) module to extract more beneficial segmentation features in multi-modal images [147].

In the task of brain tissue segmentation, Yamanakkanavar et al. performed extensive work from 2021–2022 [148–150]. In 2011, he proposed a novel brain tissue segmentation model by introducing a global attention (GA) module. The GA module can extract global context information from MR images. One year later, the authors used a lightweight multi-scale deep network integrated with a fire module for brain tissue segmentation, named SM-SegNet. In the same year, he proposed a multipath feature fusion convolutional neural network (MF2-Net) for medical image segmentation. This model was not only applied to brain tissue segmentation but also achieved better results in other regions. Both Zhai et al. and Danyananda et al. specifically focused on extracting

global contextual information [145, 151]. Zhai et al. designed an attention module with two convolutional layers, which can extract more efficient global features for brain tissue segmentation [145]. Dayananda et al. designed the global attention module to focus on global features [151]. Long et al. designed modules and constructed segmentation modules for spatial features [152]. Wu et al. introduced the graph attention network (GAT) to enhance the segmentation performance of the model [150]. Li et al. constructed a context-assisted fully attention network (CAN) by integrating 2D and 3D MR images [153]. Xiong et al. used attention modules to adjust the spatial information of the feature map to highlight the target area for small target segmentation when up-sampling [154]. Jiang et al. sequentially incorporates channel attention and self-attention mechanisms to construct a squeeze-and-excitation and self-attention (SSA) block that can effectively focus on areas requiring precise segmentation [155].

In recent years, owing to the inherent challenges of medical images, such as poor image quality, different imaging and segmentation protocols, and variations among patients, multi-domain segmentation methods have gradually become a popular topic in brain tissue segmentation [144, 156, 157]. Dorent et al. proposed a joint model that performs tissue and lesion segmentation as a unique joint task. By introducing feature channel averaging across modalities, the hetero-modal problem can be solved [144]. Similarly, Delisle et al.

used the multi-task learning method to implement the joint learning of image normalization and segmentation [156]. Kamnitsas et al. studied the unsupervised domain adaptation of adversarial networks [157]. However, despite the extraction of many multi-domain brain tissue segmentation methods, significant challenges remain to be addressed.

Discussion

Recently, deep learning has achieved a remarkable success. It has attracted great interest in many computer-related fields, including natural language processing [158], image processing [159], speech recognition [160], and bioinformatics [161]. The application of deep learning in brain tissue segmentation has undeniably propelled the field forward, offering unprecedented accuracy and efficiency compared with traditional segmentation methods. Currently, a plethora of high-quality papers on brain tissue segmentation have been published and discussed. In this research, we conducted a comprehensive analysis of more than 100 articles that focused on brain tissue segmentation through the use of deep learning techniques. We classified these articles into three categories on the basis of the age range of the subjects, namely fetuses, infants, and adults. Our findings demonstrate that the implementation of deep learning has a substantial effect on brain tissue segmentation across different age groups.

Table 1 shows most of the widely used public and challenge datasets for brain tissue segmentation. Most of these datasets are segmentations of normal brain tissue, including CSF, GM and WM, such as BrainWeb, IBSR18, IBSR20, and OASIS-1. Some datasets can also be used to study abnormal brain tissue segmentation or multi-class segmentation, such as the BrainWeb dataset, which can be used for segmentation and detection of multiple sclerosis (MS) [162]. In addition, few public datasets exist for fetal and infant brain tissue segmentation. To promote brain tissue segmentation algorithms for fetuses and infants, researchers have organized a number of segmentation challenges and released the available data, such as iSeg-2017, iSeg-2019, and FeTA 2021.

Table 2 presents a quantitative analysis of various state-of-the-art methods for fetal brain tissue segmentation via Dice. Wen et al. achieved a mean DSC of 0.9010 using the FeTA 2021 dataset, which is higher than that reported in other studies [45]. This paper introduces a new ExSwin Transformer module in deep learning. The module is designed to capture long-range dependencies and correlations between different samples, enabling the extraction of richer contextual information. The DSCs obtained via the other datasets ranged from 0.78 to 0.9. In addition, most of these methods are based on the U-Net structure to construct fetal brain tissue segmentation models [45, 47, 50, 51, 55]. The segmentation of fetal

brain tissue poses considerable obstacle because of factors such as the brain's small size, inadequate tissue contrast, and low signal-to-noise ratio. Therefore, further research is necessary to enhance the effectiveness and consistency of deep learning-based automated segmentation of fetal brain tissue, with a focus on enhancing the segmentation accuracy.

Table 3 shows the quantitative results for different infant brain tissue segmentation methods. Owing to the difficulty in acquiring infant brain MR images, many researchers have focused on two challenge datasets, iSeg-2017 and iSeg-2019. These proposed methods generally use multi-modal images combined with U-Net or DenseNet to construct segmentation models, and the mean DSC coefficient is between 0.80 and 0.96. Bui et al. proposed a multi-task learning model that incorporates an attention mechanism [56]. They achieve a DSC coefficient of 0.9790 for the segmentation task on a non-public dataset.

Many automatic brain tissue segmentation methods have been proposed for adults over the years. Tables 4, 5 and 6 briefly summarize three aspects of adult brain tissue segmentation: the network structure, prior knowledge, and attention mechanism. With the development of deep learning techniques, many mature network architectures have been developed. As soon as a new network architecture emerges, scholars will promptly apply it to their respective fields to assess its performance, as was done with CNN [86], U-Net [102], GCN [112], SegNet [107], etc. In addition, many variations or improvements of these network architectures have been applied to adult brain tissue segmentation, such as U-SegNet [97, 106], DenseNet [93], 3D Dense-U-Net [94], RP-Net [15], etc. Multi-modal image fusion strategies utilize prior knowledge, such as shape priors, learned from an assistant modality to enhance segmentation performance on a target modality [146]. This strategy is widely used in brain tissue segmentation [124–133]. To improve the interpretability of deep learning, an attention mechanism is applied to brain MR image tissue segmentation. Table 5 shows the various types of attention mechanisms used in adult brain tissue segmentation. The use of attention mechanisms in deep learning enables the identification and prioritization of the relevant information within feature maps, while effectively suppressing any irrelevant information [163]. While deep learning models excel when trained on large, carefully curated datasets, they tend to suffer from reduced performance when applied to data from unseen domains or scanners, highlighting the need for strategies such as domain adaptation and the use of larger, more diverse training datasets to ensure robustness.

Despite the numerous deep learning-based algorithms developed for brain tissue segmentation in subjects of varying ages (fetal, infant, and adult), only a limited number of them are suitable for clinical application for the following reasons:

- (1) The lack of interpretability is a significant inherent issue associated with deep learning models [164].
- (2) A general method that can be applied to brain MR images acquired from different sites and scanners is currently lacking.
- (3) Obtaining complete annotations of datasets can be a challenging and expensive task, and expert annotations may also contain errors.
- (4) In brain magnetic resonance image segmentation tasks, due to significant differences between samples, deep learning models may not have good generalizability, resulting in poor performance.

In the future, to enable the developed deep learning-based model to provide effective assistance for clinical diagnoses, we believe that the following research directions for brain tissue segmentation should be considered: (1) Integration of clinical knowledge or developing interpretable modules to enhance the interpretability of the model. (2) Construction of a large dataset containing multiple sites and multiple scanners to enhance the generalizability and robustness of the model. (3) Development of unsupervised or semi-supervised models to overcome the constraints of expert annotations.

Conclusion

In this survey, we present an exhaustive study of various existing deep learning-based methods for fetal, infant, and adult brain tissue segmentation. Deep learning technology has proven to be a powerful tool in automatic brain tissue segmentation, leading to significant success across various age groups. We structurally categorized and summarized the deep learning-based segmentation methods. Furthermore, we propose three reasons for the limited clinical application. However, this review still has the following limitations: (1) it focuses only on segmentation methods for brain tissue; (2) it focuses only on deep learning-based brain tissue segmentation methods. In the future, we will summarize more comprehensive segmentation methods for MRI brain images, such as for brain tissue, and brain tumors. In addition, we will add methods other than deep learning to comprehensively help researchers understand the development dynamics of MR brain images.

Author contributions Liang Wu and Shirui Wang: Conceptualization, Writing of first draft. Jun Liu and Lixia Hou: Data Resources. Liang Wu and Li Song: Methodology, Model Development and Validation. Weizhao Lu, Jianfeng Qiu, and Ming Zhang: Methodology, Review and editing of manuscript. Peng Zhao, Li Song, and Na Li: Conceptualization, Methodology, Review and editing of manuscript.

Funding This study was financially supported by the Taishan Scholars Program of Shandong Province (grant no. TS201712065), Academic Promotion Program of Shandong First Medical University (grant no.

2019QL009), Science and Technology funding from Jinan (grant no. 2020GXRC018), Science and Technology Innovation Development Project of Tai'an City (grant no. 2022NS331, 2023NS381, and 2023NS145), the Project of Shandong Province Youth Innovation Team (grant no. 2023KJ182), Shandong Provincial Medical and Health Science and Technology Program (grant no. 202325020521) and Natural Science Foundation of Inner Mongolia (grant no. 2022LHMS06005).

Data availability The authors confirm that the data supporting the findings of this study are available within the article. And all relevant data are within the paper.

Declarations

Conflict of interest The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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